

# Mutation Zone Handoff Note

Audience: pixel artist using `template-editor-v4.html`

## What a mutation zone is

A mutation zone is a rectangular area where the game is allowed to slightly change the template when it generates visual variants.

- Inside a zone, the game may add pixels, remove pixels, shift a small area, thicken outlines, add patterns, or add small detail clusters.
- Outside a zone, the template stays exactly as drawn.
- Variant 0 is always the untouched original. Mutations only happen on the generated variants.

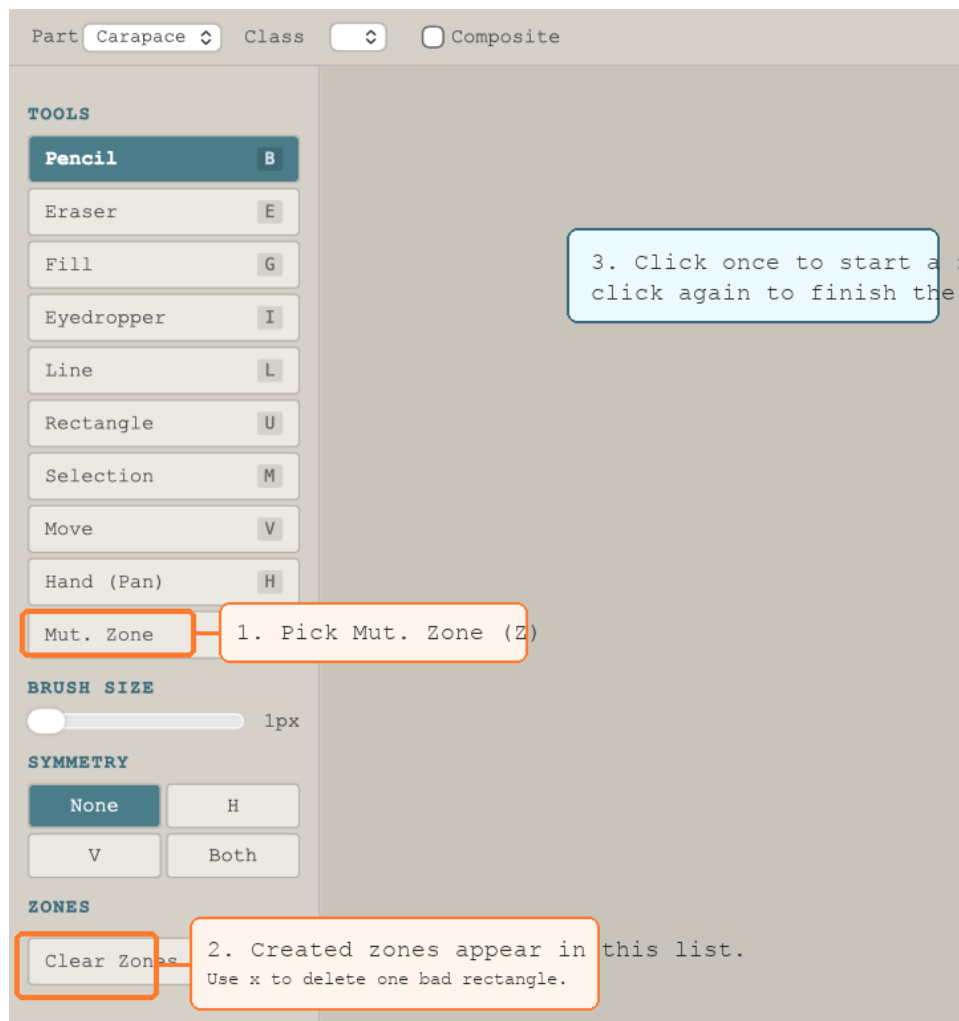
The safest mental model is:

mutation zone = permission to vary this area

Not:

mutation zone = important detail area

## How to draw one in the editor



Mutation zone tool overview

1. Pick Mut. Zone in the left toolbar, or press Z.
2. Click once on the canvas to start the rectangle.

3. Click again to finish the rectangle.
4. The zone will appear in the **Zones** list.
5. Use the **x** in that list to remove a bad zone.

## Current state of the Apex workspaces

The current files in `packages/asset-gen/apex_characters_clwb` are already useful as reference:

- they are composite full-character workspaces
- they currently have 0 mutation zones defined
- each workspace has 6 body-part layers



Current Apex designer workspaces

This means the handoff is not asking the designer to redraw anything. It is asking them to decide where variation is safe on top of the work they already finished.

## What makes a good zone

Use zones on ornamental or flexible areas:

- shell ridges
- texture bands
- tips, spikes, serrations
- decorative markings
- flat interior areas that can take dots/stripes/checkerboard

Keep these stable:

- the main silhouette
- anchor / attachment area
- joints and hinges
- eyes, pupils, and tiny precision details
- narrow connection points

Rule of thumb:

If a 1-pixel shift would break the shape, do not zone it.

## Recommended zone plan per part

Aim for 2-4 small zones, not one giant zone.

- **Zone 1: edge zone** Place it on an outer edge, tip, or ridge that can tolerate tiny silhouette changes.
- **Zone 2: interior zone** Place it on a flatter interior area that can take patterns or overlays.
- **Zone 3: detail zone** Place it on a secondary feature that can shift slightly without changing the identity of the part.
- **Zone 4: optional** Only if the body part is large enough and has another clearly separate feature.

## Composite-mode note for the current files

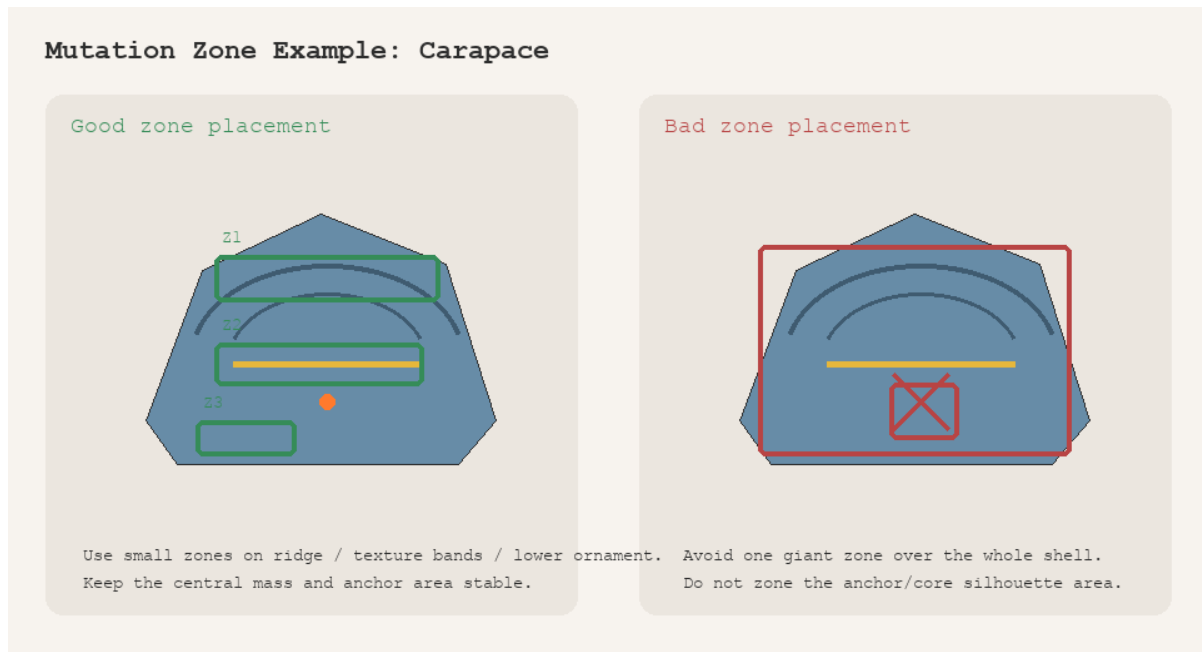
The Apex .clwb files are composite workspaces, so zones are drawn on the full 64×64 character canvas.

That means:

- draw the rectangle over the actual body part area you want to vary
- a zone over the claw region should only affect claw pixels, because the other parts do not have pixels there
- avoid drawing broad rectangles that overlap multiple body parts unless you intentionally want all of them to be eligible for mutation there

## Good vs bad examples

### Carapace



Carapace good vs bad

Good:

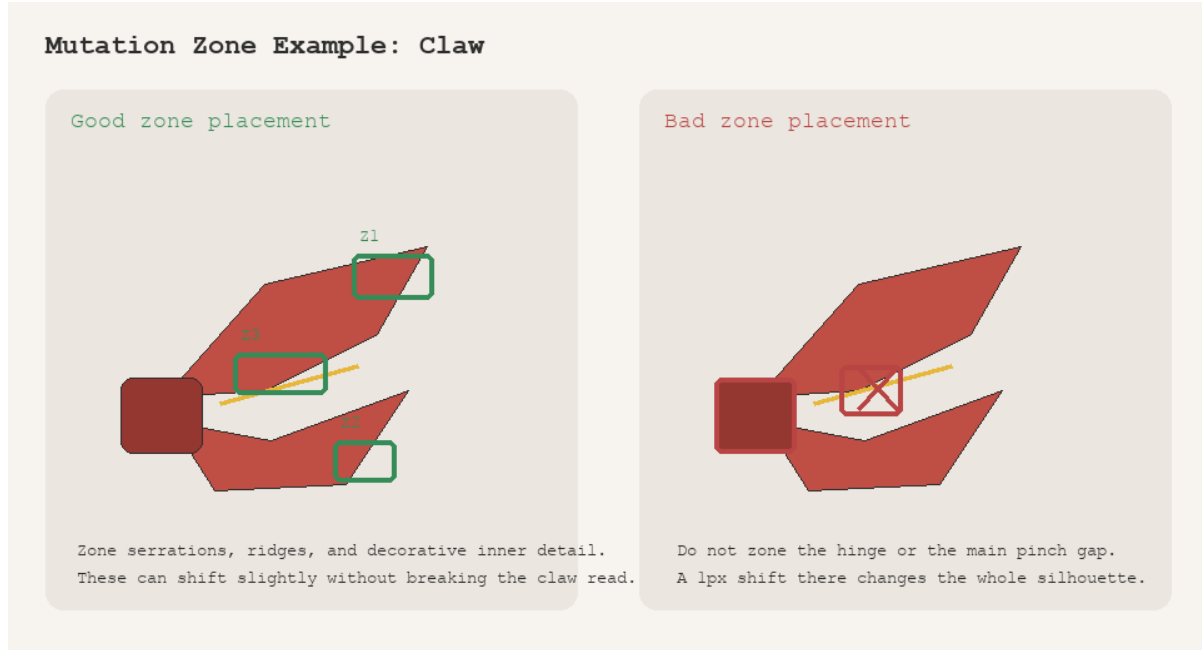
- small zones on ridge, stripe band, and lower ornament
- stable center mass
- stable anchor/core area

Bad:

- one giant zone over most of the shell

- zoning the anchor/core silhouette area

## Claw



Claw good vs bad

Good:

- zone serrations, ridges, and decorative inner detail
- keep the main pinch shape intact

Bad:

- zoning the hinge
- zoning the pinch gap
- zoning any place where a 1-pixel move changes the whole claw read

## Important current tool limitation

Right now, every new zone is created with all six mutation behaviors enabled by default.

That means a zone currently says:

**any supported mutation is acceptable here**

So when placing zones, be conservative. Only mark areas that can survive all of the following:

- slight silhouette growth
- slight silhouette loss
- small 1-pixel shifts
- thicker outline
- interior patterning
- tiny decorative overlays

## Quick checklist before export

Before export, confirm all of the following: 2-4 small zones, some stable area left untouched, zones kept away from the anchor/attachment area, the main identity silhouette left stable, and every zone able to survive a 1-pixel change.